

Chapter 1

Introduction

1.1 Objectives of the Internship

The primary objective of this internship at ISRO ISTRAC is to facilitate a deep integration of advanced mathematical communication theory with the practical complexities of satellite navigation. Academically, the goal is to master the implementation of Galois Field ($\text{GF}(2)$) arithmetic and the properties of Primitive Polynomials to generate robust pseudo-random noise (PRN) sequences. While theoretical knowledge provides the foundation, this internship aims to apply these concepts to Linear Feedback Shift Registers (LFSR) to solve real-world signal interference issues. By utilizing AI techniques for sequence optimization, the academic focus shifts toward reducing the computational complexity of traditional exhaustive search methods, thereby identifying sequences with superior auto-correlation and cross-correlation properties. This research is critical for improving the signal acquisition time and overall accuracy of Global Navigation Satellite System (GNSS) applications, particularly for the NavIC constellation. On a personal level, this internship is designed to cultivate a professional scientific temper within the framework of India's premier space agency. Working at ISTRAC provides a unique opportunity to witness the lifecycle of mission-critical software and hardware development. One of my core personal goals is to enhance my problem-solving capabilities by tackling the "Near-Far" problem in satellite communications through innovative AI-driven approaches. Furthermore, I aim to develop the soft skills required in high-stakes engineering environments, such as technical documentation, collaborative research, and navigating the rigorous quality standards maintained by ISRO. By the conclusion of this tenure, I intend to have bridged the gap between being a student and a contributing researcher in the field of satellite-based navigation and timing services.

1.2 Overview of the Organization

The ISRO Telemetry, Tracking and Command Network (ISTRAC), headquartered in Bangalore, serves as the essential ground segment backbone for the Indian Space Research Organisation. Since its inception, ISTRAC has evolved from a series of individual tracking stations into a sophisticated, unified network that supports a vast array of space missions. Its primary mission is to provide telemetry, tracking, and command (TTC) services for ISRO's Low Earth Orbit (LEO) satellites, launch vehicle missions, and interplanetary expeditions. This organization was instrumental in the success of landmark missions such as the Mars Orbiter Mission (Mangalyaan) and the Chandrayaan

series. Beyond tracking, ISTRAC is the operational heart of the Navigation with Indian Constellation (NavIC) and the GPS Aided GEO Augmented Navigation (GAGAN) systems, which provide independent and accurate positioning services to the Indian subcontinent and its surrounding regions. The organization operates at a significant scale, maintaining a global network of ground stations in locations such as Port Blair, Lucknow, Mauritius, Brunei, and Biak (Indonesia). This allows for near-continuous visibility of spacecraft, which is vital for monitoring satellite health and executing precise orbital maneuvers. ISTRAC operates within the government-led aerospace and defense industry, employing thousands of top-tier scientists and engineers dedicated to achieving indigenous technological sovereignty. The organizational mission is rooted in the "Atmanirbhar Bharat" (Self-Reliant India) initiative, focusing on developing homegrown solutions for satellite-based navigation, search and rescue services, and deep space communication.

Core Mission Responsibilities

1. **Mission Control (MOX):** The Mission Operations Complex (MOX) in Bengaluru is the nerve centre where health analysis, orbit determination, and satellite maneuvers are executed.
2. **Navigation (NavIC):** ISTRAC manages the ground segment for the Navigation with Indian Constellation (NavIC), maintaining the high-precision atomic clocks and 21 ranging stations required for independent regional positioning.
3. **Space Situational Awareness (IS4OM):** The ISRO System for Safe and Sustainable Space Operations Management, inaugurated at ISTRAC, monitors space debris to safeguard Indian assets from potential collisions.
4. **Search and Rescue (SAR):** As part of an international programme (COSPAS-SARSAT), ISTRAC provides satellite-aided search and rescue services by processing distress signals from across the region.

1.3 Internship Role/Position

During my internship at ISTRAC, I am assigned to the Navigation Systems Division, where I function as a Technical Intern specializing in Signal Design and Optimization. My specific role involves the "Optimization of Sequence Selection for GNSS Applications using AI Techniques." In the context of satellite navigation, the choice of PRN sequences is paramount; these codes must be nearly orthogonal to one another to prevent mutual interference while allowing the receiver to lock

onto the signal even in low-signal-to-noise ratio environments. My work revolves around the mathematical modeling of these sequences using LFSR configurations. I am responsible for analyzing various Primitive Polynomials within a specific Galois Field to ensure that the generated sequences are of maximum length (m-sequences) and possess the requisite randomness properties necessary for secure and reliable transmission. The AI component of my role involves developing and implementing optimization algorithms—such as Genetic Algorithms or Neural Networks—to evaluate millions of possible sequence combinations. Unlike traditional methods that rely on pre-defined Gold or Weil codes, my objective is to use AI to discover "optimal" sequences that minimize the peak sidelobe levels of the cross-correlation function. This is a critical task for enhancing the NavIC system's performance in "urban canyon" environments where multipath interference is a major challenge. My daily responsibilities include performing software simulations to validate the correlation properties of new sequences and documenting the performance gains compared to existing standards. Working within this specialized department, I am gaining firsthand experience in how AI can be leveraged to refine the physical layer of satellite communication, contributing directly to the modernization of India's indigenous navigation capabilities.

Chapter 2

About the Organization

2.1 Organization Overview

The Indian Space Research Organisation (ISRO) Telemetry, Tracking and Command Network, commonly known as **ISTRAC**, is a premier public sector unit under the Department of Space, Government of India. Headquartered in **Bangalore, Karnataka**, ISTRAC serves as the ground segment backbone for India's space endeavors. Established in the early 1970s, the organization was created with the specific mandate of providing integrated tracking support for all of ISRO's low-earth orbit (LEO) satellites and launch vehicle missions. Over the decades, it has evolved from a few scattered ground stations into a sophisticated global network. ISTRAC's history is deeply intertwined with India's rise as a space power, having provided the critical communication links for historic milestones such as the Aryabhata satellite launch, the Mars Orbiter Mission (Mangalyaan), and the Chandrayaan lunar missions. Today, it operates as a specialized hub that blends aerospace engineering with advanced computer science to ensure that India maintains a constant, reliable connection with its assets in space.



Fig:2.1 ISRO

2.2 Vision, Mission, and Objectives

The vision of ISTRAC is to achieve excellence in ground segment operations and to provide world-class telemetry, tracking, and command services for the sustainable exploration of outer space. Its mission is centered on the development and maintenance of a robust ground station network that can support complex interplanetary missions, satellite navigation, and real-time launch vehicle tracking. Strategically, ISTRAC focuses on "Technological Sovereignty," ensuring that India possesses indigenous capabilities for satellite-based services without relying on foreign entities. The key objectives include maintaining 99.9% uptime for satellite links, managing the Indian Regional Navigation Satellite System (NavIC), and expanding the deep space network to support future manned missions like Gaganyaan. By prioritizing indigenous innovation, ISTRAC aims to reduce

the cost of space operations while increasing the precision and reliability of timing and positioning data provided to both .

2.3 Products and Services

Space Programme. Its primary "product" is the **Navigation with Indian Constellation**



Figure 2.2 : 32m diameter antenna- for MOM

independent regional navigation satellite system that provides accurate real-time positioning and timing services over India and a region extending 1,500 km around it. Additionally, ISTRAC provides **Mission Operations Services**, which involve the continuous monitoring of satellite health and the execution of tele-command maneuvers to maintain orbital paths. The organization also operates the **Indian Space Science Data Center (ISSDC)**, which serves as the primary archive for all data generated by Indian scientific missions. Its target "customers" range from the general public (using NavIC for mobile navigation) to the Indian Armed Forces, as well as international space agencies that utilize ISTRAC's tracking stations for global cross-support. Through its Search and Rescue (SAR) services, ISTRAC also provides critical distress alert detection for maritime and safety designed to support deep-space missions such as Mars Orbiter Mission (MOM).

For MOM, the 32 m antenna played a critical role in transmitting telecommands and receiving telemetry strength at such vast distances, a large dish diameter was essential to achieve higher gain and better signal-to-noise ratio.

The antenna operates in S-band and X-band frequencies, enabling reliable communication, precise tracking, and accurate orbit determination. It also supports ranging and Doppler measurements required for trajectory corrections. The successful performance of this antenna during MOM demonstrated India's capability in deep-space communication and strengthened ISRO's infrastructure for future interplanetary missions.

ISRO provides a wide range of products and services that support national development, scientific research, and global space collaboration. Its major products include communication satellites, remote sensing satellites, navigation satellites, scientific spacecraft, and launch vehicles. Satellite systems developed by ISRO enable television broadcasting, telemedicine, tele-education, disaster management, weather forecasting, and precise navigation services.

2.4 Organizational Structure

ISTRAC follows a well-defined, multi-tiered technical hierarchy designed to manage the complexities of global space operations. At the pinnacle is the Director, followed by Associate Directors who oversee major divisions such as Mission Operations, Ground Station Networks, and Navigation Systems. The organization is divided into specialized departments, including the **Navigation Systems Division**, Data Processing, and Software Development. Within these departments, teams are structured by technical expertise, where senior scientists lead groups of engineers and researchers. My internship is situated within the Navigation Systems Division, which is specifically responsible for the ground segment of the NavIC constellation. This department plays a pivotal role in the organization by ensuring that the signals transmitted by satellites are accurate and secure. It acts as the bridge between raw satellite data and the usable positioning information used by end-users, making it one of the most mathematically intensive units within ISTRAC.

2.5 Work Environment and Culture

The work culture at ISTRAC is defined by a rigorous "Scientific Temper" and a commitment to precision. As a mission-critical facility, the environment is one of disciplined professionalism where technical accuracy is paramount. The organization fosters a collaborative atmosphere where senior scientists often mentor interns and junior engineers, encouraging an open exchange of ideas, especially in emerging fields like the application of AI in signal processing. Ethics and data integrity are at the core of their professional practices, as the data handled has national security implications. For interns, ISTRAC maintains a supportive yet demanding policy; we are expected to adhere to the organization's strict security protocols and working hours while being given the intellectual freedom to explore innovative solutions to complex problems. This culture of excellence ensures that every individual, whether a permanent staff member or an intern, understands their contribution toward India's broader goals of space exploration and national self-reliance.

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Chapter 3

Objectives of Internship

3.1 Purpose of the Internship

The decision to undertake this internship at ISRO ISTRAC was driven by a combination of academic necessity and a deep-seated interest in satellite communication systems. Primarily, this internship fulfills the mandatory academic requirement of my degree program, providing a bridge between classroom theory and industrial application. However, beyond the curriculum, the fundamental purpose was to gain firsthand exposure to the space industry, which operates at a level of precision and scale far beyond typical academic labs. By working on the "Optimization of Sequence Selection for GNSS," the internship allowed me to transition from abstract mathematical modeling to solving real-world engineering problems. The objective was to understand the lifecycle of a technical project within a government space agency, observing how theoretical concepts like signal processing are translated into mission-critical services that support millions of users across the Indian subcontinent.

3.2 Learning Objectives

- **Technical Proficiency:**One of the major objectives of the internship was to develop technical skills related to Artificial Intelligence and Machine Learning. The internship focused on improving knowledge in areas such as data analysis, data preprocessing, statistical techniques, and data visualization. Practical exposure to tools and technologies such as Python, SQL, Git, and visualization libraries helped in strengthening technical understanding.
- **Soft Skills Development:**The internship also aimed to improve professional and interpersonal skills. Regular assignments and documentation activities helped in improving communication skills, report writing, analytical thinking, and time management. Working on structured tasks also encouraged discipline, consistency, and problem-solving abilities.
- **Domain Knowledge:**Another important objective was to gain knowledge about Artificial Intelligence concepts and their practical applications. The internship provided exposure to machine learning workflows, neural networks, cognitive science concepts, ethical AI practices, and real-world AI use cases in different industries.
- **Academic Alignment:**The internship was closely aligned with the Computer Science and Engineering curriculum. The concepts covered during the internship complemented academic subjects such as Machine Learning, Artificial Intelligence, Database Management Systems, Data Analytics, and Software Engineering. This helped in strengthening the practical understanding of concepts learned during academics.

3.3 Career Relevance

This internship is a cornerstone of my professional development, aligning perfectly with my career aspirations in the fields of Aerospace Engineering and Applied Artificial Intelligence. Having "ISRO ISTRAC" on my professional profile significantly enhances my employability, as it demonstrates my ability to work within a highly regulated, mission-critical environment. The specific skills I am acquiring—such as optimizing sequences for GNSS—are highly sought after in the telecommunications, defense, and autonomous vehicle industries. As the world moves toward more precise positioning requirements for IoT and self-driving technology, my expertise in PRN code optimization will be a significant asset. This experience not only validates my technical capabilities but also showcases my ability to handle complex mathematical challenges, positioning me as a strong candidate for future roles in R&D or systems engineering within the global space and tech sectors.

3.4 Organizational Objectives

From the perspective of ISTRAC, the internship was designed to bring a fresh, research-oriented approach to a persistent challenge in GNSS signal design. The organization's objective was to explore the feasibility of using AI and Machine Learning to automate and optimize the selection of spreading codes, a task that has traditionally been labor-intensive and computationally expensive. As an intern, I was expected to contribute by conducting extensive simulations and benchmarking the performance of various primitive polynomials against existing standards like Gold Codes. My contribution lies in providing a data-driven analysis that could potentially lead to better signal acquisition and tracking performance for future iterations of the NavIC system. By integrating my academic knowledge of AI with their established expertise in navigation, the organization aims to identify more efficient ways to minimize signal .

Another key objective is to strengthen India's capabilities in human spaceflight, interplanetary missions, and advanced scientific exploration. Through initiatives such as Gaganyaan, lunar and solar missions, and upcoming planetary explorations, ISRO seeks to expand scientific knowledge and technological innovation. Additionally, ISRO promotes research, capacity building, and international collaboration to enhance global cooperation in peaceful uses of outer space.

Overall, ISRO's objective is to integrate space technology into societal progress, economic development, and scientific advancement while maintaining cost-effective and sustainable mission execution.

3.5 Scope of Work

The scope of my work at ISTRAC is precisely defined within the boundaries of the

NavigationSystemsDivision. The focus area is restricted to the physical layer of GNSS signals, specifically the generation and optimization of PRN sequences. The work involves a deep dive into the mathematical properties of sequences generated by shift registers, focusing on finding those with the lowest cross-correlation sidelobes. The domain is strictly limited to the optimization of sequences for GNSS applications and does not extend to the orbital mechanics or hardware fabrication aspects of the satellites. Key tasks involve the simulation of LFSR structures, the implementation of AI-based search algorithms, and the comparative analysis of correlation results. By narrowing the scope to these specific areas, the internship ensures a high degree of technical depth, allowing for a concentrated effort on improving the signal quality .

Chapter 4

Internship Activities and Tasks

4.1 Tasks and Responsibilities

During the internship at ISTRAC within the Navigation Systems (NSA) department, the workload was structured into progressive phases of research, mathematical modeling, and technical implementation. These responsibilities were categorized to ensure a comprehensive understanding of the NavIC navigation system and the underlying signal processing required for GNSS applications.

Research and Domain Study

The initial phase involved an intensive study of the **NavIC (Navigation with Indian Constellation)** architecture. This included researching the three-segment architecture (Space, Control, and User segments) and understanding how they synchronize to provide Precise Point Positioning (PPP). Further research was conducted into dual-band frequency transmissions (L5 and S-band), which are critical for correcting ionospheric delays more effectively than single-band systems.

Technical Work and Algorithm Implementation

The core technical responsibility focused on the generation and optimization of **Pseudo-Random Noise (PRN)** codes, which serve as the "fingerprints" for individual satellites. This involved:

- **LFSR Implementation:** Utilizing Boolean logic and XOR operations to design and analyze Linear Feedback Shift Registers (LFSR).
- **Sequence Optimization:** Applying primitive polynomials to ensure shift registers reached their "Maximum Length Sequence" for hardware efficiency.
- **Gold Code Generation:** Combining LFSR outputs to create code libraries with high auto-correlation for signal locking and low cross-correlation to prevent signal "crosstalk".
- **Simulation:** Using Python to implement and test these navigation signal algorithms.

On the technical front, implementing LFSRs using Boolean logic and primitive polynomials offered practical insight into sequence generation theory and finite field mathematics. The design and optimization of Maximum Length Sequences improved understanding of hardware-efficient coding structures. Furthermore, generating Gold codes demonstrated the importance of achieving strong auto-correlation for precise signal acquisition and low cross-correlation for multi-satellite environments. Python-based simulations validated theoretical models and ensured algorithmic

correctness. Overall, the internship successfully integrated research, mathematical modeling, and software implementation into a comprehensive GNSS-focused learning experience. the internship at ISTRAC within the Navigation

Systems (NSA) department provided a structured and in-depth exposure to both theoretical and practical aspects of satellite navigation. The progressive workflow—from domain research to algorithm implementation—enabled a clear understanding of how the NavIC system ensures accurate and reliable positioning services. Studying the three-segment architecture and dual-band transmission strategy strengthened conceptual clarity on error correction mechanisms such as ionospheric delay mitigation and Precise Point Positioning (PPP).

Navigation with Indian Constellation

The initial phase of the internship was dedicated to building a strong theoretical foundation in the NavIC (Navigation with Indian Constellation) system and its operational framework. A detailed study was conducted on the three-segment architecture—Space Segment (satellites), Control Segment (ground monitoring and control stations), and User Segment (receivers). Emphasis was placed on understanding how precise time synchronization, orbital data updates, and signal transmission collectively enable accurate positioning services. Special focus was given to Precise Point Positioning (PPP) and how correction parameters are integrated to enhance positioning accuracy.

Further domain research explored NavIC's dual-frequency transmission strategy in the L5 and S-bands. The study highlighted how dual-band signals significantly reduce ionospheric delay errors compared to single-frequency GNSS systems. This phase strengthened conceptual clarity in satellite navigation principles, signal propagation, timing accuracy, and atmospheric error mitigation techniques essential for reliable navigation performance.

Technical Work and Algorithm Implementation

The technical phase centered on the design and optimization of Pseudo-Random Noise (PRN) codes, which uniquely identify each satellite signal. Linear Feedback Shift Registers (LFSRs) were implemented using Boolean logic and XOR operations, providing hands-on exposure to digital sequence generation. Primitive polynomials over finite fields were applied to ensure generation of Maximum Length Sequences (m-sequences), achieving optimal periodicity and hardware efficiency. Gold code generation was performed by combining outputs of two carefully selected LFSRs. This ensured strong auto-correlation properties for precise signal acquisition and low cross-correlation

Artificial Intelligence concepts practically. These tools supported learning, data analysis, visualization, and project development activities.

Programming Languages

- **Python** – Used for understanding Machine Learning concepts, data analysis, preprocessing, and AI-related tasks.

Libraries and Frameworks

- **NumPy** – Used for numerical computations and array operations.
- **Pandas** – Used for handling and analyzing datasets.
- **Matplotlib and Seaborn** – Used for data visualization and graphical analysis.
- **Scikit-learn** – Used for Machine Learning model understanding and implementation.

Development and Collaboration Tools

- **Git** – Used for version control and progress tracking.
- **Pycharm** – Used as the primary development environment for coding and project implementation.

AI Platforms and Tools

- **MidJourney** – Used for generating AI-based creative visuals and experimenting with artistic styles.

4.2 Projects Involved In

1. PRN Code Generation and Optimization

This project concentrated on the systematic design and validation of Pseudo-Random Noise (PRN) codes used for satellite identification in navigation systems.

- Designed Linear Feedback Shift Registers (LFSRs) using Boolean logic and XOR-based feedback mechanisms.
- Selected appropriate primitive polynomials over GF(2) to generate Maximum Length Sequences (m-sequences).
- Verified periodicity properties ($2^n - 1$) and statistical balance of generated sequences.
- Performed auto-correlation analysis to ensure sharp peaks for precise signal acquisition.

- Evaluated cross-correlation properties to minimize inter-satellite interference.
- Implemented Python simulations to automate sequence generation, testing, and validation.
- Compared multiple feedback configurations to improve hardware efficiency and reliability.

This work strengthened practical understanding of sequence theory, finite field mathematics, and GNSS signal acquisition principles.

2. $\mathbb{Z}_4$ Sequence Generation using LFSR

To extend beyond binary sequence models, advanced sequence generation was carried out over the ring $\mathbb{Z}_4$ (integers modulo 4).

- Implemented LFSR structures operating under modulo-4 arithmetic.
- Developed recurrence relations for $\mathbb{Z}_4$ -based feedback systems.
- Analyzed state transitions and orbit structures in $\mathbb{Z}_4$ algebra.
- Studied balance properties and distribution of symbols (0,1,2,3).
- Evaluated sequence periodicity and structural behavior.
- Compared correlation characteristics with traditional binary sequences.
- Explored potential advantages of non-binary sequences in improving flexibility of code design.

This project deepened understanding of algebraic structures, non-binary coding systems, and their relevance in navigation signal design.

3. Threshold-Based Sequence Selection

A systematic threshold analysis approach was implemented to select optimal sequences from large candidate sets.

- Computed cross-correlation values among generated sequences.
- Defined acceptable threshold limits based on GNSS performance requirements.
- Filtered sequences that satisfied predefined correlation constraints.
- Analyzed trade-offs between threshold strictness and available sequence pool size.
- Validated selected sequences through simulation and performance testing.

This ensured the final sequence library maintained strong auto-correlation for signal locking and sufficiently low cross-correlation for multi-satellite operation.

4. NavIC

NavIC (Navigation with Indian Constellation), also known as IRNSS (Indian Regional Navigation Satellite System), is India's independent regional satellite navigation system developed by the Indian

Space Research Organisation (ISRO) to provide accurate position, velocity, and time information over India and a region extending about 1500 kilometers beyond its borders. It was established to ensure strategic autonomy and reduce dependence on foreign navigation systems such as GPS, particularly for critical applications like defense, disaster management, transportation, and national infrastructure. NavIC operates on the fundamental principle of satellite-based trilateration, where multiple satellites transmit signals containing precise timing and orbital information, and a receiver calculates its position by measuring the time taken for these signals to travel from satellites to the receiver. Since radio signals move at the speed of light, even nanosecond-level timing errors can cause significant positioning inaccuracies, which is why NavIC satellites are equipped with highly stable atomic clocks. The system follows a three-segment architecture: the space segment, control segment, and user segment. The space segment consists of a constellation of satellites placed in geostationary (GEO) and geosynchronous (GSO) orbits, ensuring continuous visibility over the Indian region and stronger signal availability compared to systems that rely only on medium Earth orbit satellites. The control segment includes ground stations responsible for tracking satellites, monitoring their health, synchronizing onboard clocks, computing orbital parameters, and uploading correction data to maintain accuracy and reliability. The user segment comprises receivers integrated into devices such as smartphones, vehicle navigation systems, maritime equipment, aviation instruments, and surveying tools that process NavIC signals to determine precise location information. One of the most distinctive features of NavIC is its dual-frequency transmission capability, operating in the L5 band (1176.45 MHz) and the S-band (2492.028 MHz). While most global navigation systems operate primarily in the L-band, NavIC's inclusion of the S-band provides improved resistance to atmospheric disturbances and enhances signal reliability. The dual-frequency design plays a crucial role in correcting ionospheric delay, which is one of the largest sources of error in satellite navigation. Because ionospheric delay affects signals differently depending on frequency, transmitting signals at two different frequencies allows receivers to calculate and compensate for this error more accurately, improving positioning precision compared to single-frequency systems. NavIC provides two primary services: the Standard Positioning Service (SPS), which is open and available for civilian use, and the Restricted Service (RS), which is encrypted and intended for authorized users, particularly defense and strategic applications. The SPS supports general navigation needs such as vehicle tracking, fleet management, maritime navigation, agriculture, surveying, and disaster response, while the RS offers enhanced accuracy and secure communication for military operations and sensitive missions. In terms of performance, NavIC is designed to deliver position accuracy better than 20 meters across India, and with advanced receiver technologies and correction techniques such as Precise Point Positioning (PPP),

even higher accuracy can be achieved. Like all satellite navigation systems, NavIC performance can be affected by factors such as satellite geometry, atmospheric conditions, multipath reflections from buildings, and receiver quality. However, continuous monitoring and real-time correction updates from the control segment help mitigate these issues. NavIC has significant applications in transportation systems, including road navigation, railway safety, and intelligent transport systems; in maritime operations, including coastal navigation and fishing vessel tracking; in disaster management for coordinating rescue and relief efforts; in agriculture for precision farming; and in infrastructure development and surveying. The development of NavIC represents a major technological milestone for India, showcasing capabilities in satellite design, launch operations, orbital mechanics, atomic clock integration, and advanced signal processing. Beyond civilian benefits, NavIC strengthens national security by ensuring that critical navigation services remain available even during geopolitical disruptions that might restrict access to foreign systems. Furthermore, NavIC is increasingly being integrated into consumer electronics, including smartphones and vehicle navigation units, promoting wider public adoption. As satellite navigation continues to evolve with multi-constellation interoperability, NavIC is designed to work alongside other GNSS systems to enhance availability and accuracy in combined receivers. Future enhancements may include expanding the satellite constellation, improving clock stability, and broadening coverage beyond the current regional footprint. Overall, NavIC is a robust, reliable, and strategically significant satellite navigation system that supports India's technological independence while providing essential positioning services for civilian, commercial, and defense applications.

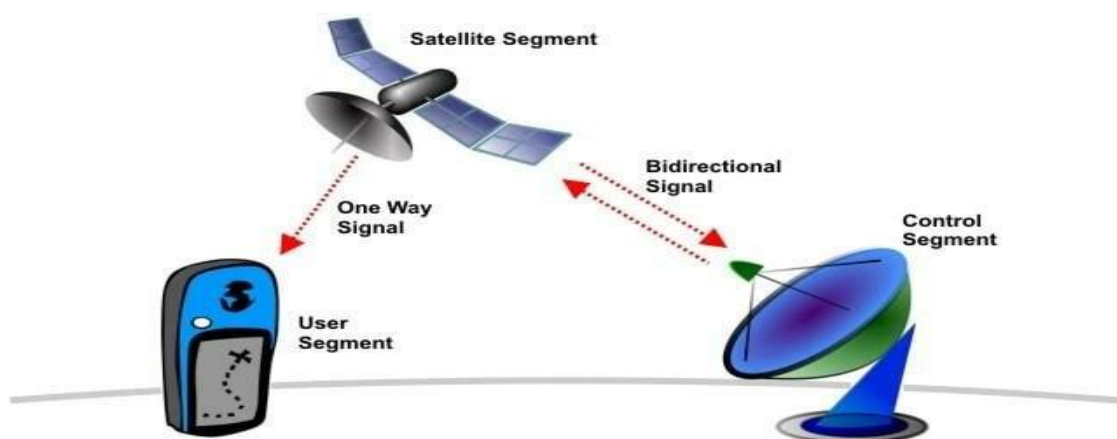


Fig:4.1 Satellite Working

4.3 Detailed Research Contribution: The Modified Clique Algorithm

The research undertaken during this internship at ISTRAC addressed a fundamental challenge in the selection of spreading sequences for modern satellite navigation constellations. In the context of the Navigation with Indian Constellation (NavIC) and emerging low earth orbit mega-constellations, the traditional reliance on algebraic families has become insufficient for meeting strict interference requirements. While modern literature has heavily favored heuristic and machine learning approaches to navigate the massive combinatorial search spaces of code pools, these methods possess an inherent flaw. They typically optimize for average correlation metrics, which allows for the emergence of rogue sequence pairs that violate hard cross-correlation thresholds. Such violations can lead to localized receiver degradation, causing cycle slips or a total loss of signal lock. This work proposed a significant paradigm shift by formulating the sequence set selection process as a graph-theoretic problem.

The technical core of the project involved the implementation of the Modified Clique Enumeration algorithm to guarantee absolute fidelity in signal compatibility. By modeling a universe of candidate spreading sequences as an undirected graph, each sequence was mapped to a unique vertex. An edge between any two vertices was only established if the pairwise cross-correlation between the sequences remained strictly below a hard interference bound defined by system designers. Within this framework, a clique—which is a subset of vertices where every two distinct vertices are adjacent—represents a perfectly compatible set of sequences. The algorithm specifically seeks maximal cliques, which are sets that cannot be expanded by adding further vertices without violating the defined threshold. This exhaustive combinatorial approach provides a complete catalog of valid sequence allocations, ensuring that no pair in the selected set exceeds acceptable interference limits.

The Modified Clique Enumeration algorithm operates in two distinct phases to ensure mathematical certainty in its output. In the first phase, it constructs a valid edge set by parsing correlation logs and filtering pairs based on the specified threshold. The second phase systematically evaluates combinations of vertices to identify all valid cliques, followed by a pruning phase that removes non-maximal sets. While the exhaustive search approach carries a high worst-case time complexity, it proved to be highly practical for orbital shell micro-allocation at ISTRAC. By pre-filtering global pools into localized subsets, the algorithm provides engineers with discrete, mathematically guaranteed non-interfering code banks. This implementation establishes the algorithm as a gold-standard benchmark for the industry, effectively proving the optimality gaps of heuristic models

Beyond the implementation of the algorithm itself, the research explored how these exact combinatorial methods outperform probabilistic optimization. By ensuring a deterministic and reproducible result that is independent of random initialization, the Modified Clique Enumeration approach allows for the definition of strict quality of service limits in satellite tracking. This is particularly critical as satellite navigation evolves toward denser constellations where the tolerance for statistical anomalies drops to zero. The work performed during this internship successfully bridged the gap between abstract graph theory and practical satellite communication, providing a structured and reliable methodology for optimizing navigation signals. This research contribution not only enhances the current performance of the NavIC ground segment but also provides a foundational framework for future research into high-fidelity sequence selection in the aerospace domain.

The initial phase involved a deep dive into the NavIC architecture, where I mastered its three-segment structure: Space, Control, and User segments. Understanding how these segments synchronize for Precise Point Positioning (PPP) was essential for recognizing the importance of signal integrity. A significant learning outcome was the analysis of dual-band frequency transmissions in the L5 and S-bands, which allow NavIC to correct ionospheric delays more effectively than traditional single-band systems. This highlighted how the quality of pseudo-random noise (PRN) codes—the "fingerprints" of satellites—directly impacts national technological sovereignty. The core technical contribution was the development of a graph-theoretic paradigm for sequence selection. In modern LEO mega-constellations, the tolerance for signal correlation anomalies is nearly zero. While machine learning often optimizes for "average" metrics, it frequently allows "rogue pairs" that violate strict cross-correlation thresholds. My work addressed this by formulating sequence selection as a Correlation Threshold Graph problem, where vertices represent candidate sequences and edges represent compatibility. This allowed the use of exact combinatorial methods to identify sequence sets with a 100% mathematical guarantee of zero threshold violations. The implementation of the Modified Clique Enumeration (MCE) algorithm stood as the central achievement. Unlike probabilistic models, the MCE algorithm performs an exhaustive combinatorial search to identify maximal cliques—sets that cannot be expanded without violating interference constraints. To validate this framework, I conducted simulations in Python comparing MCE results against Genetic Algorithm (GA) baselines. The results revealed that while GAs converge to high average performance, they often overlook individual pair violations that MCE successfully eliminates.

This established the MCE algorithm as a "gold-standard benchmark" for evaluating the optimality

4.3 Course Abstract (NSDC/SEC)

The internship program focused on providing practical exposure to Artificial Intelligence, Machine Learning, and modern AI-based tools. The course was designed to help students understand both theoretical concepts and real-world applications of AI technologies.

The course covered important topics such as:

- Fundamentals of Artificial Intelligence and Machine Learning
- Learning paradigms including supervised, unsupervised, and reinforcement learning
- Data preprocessing and visualization techniques
- Neural Networks and Cognitive Science concepts
- AI ethics and responsible AI practices
- AI-generated creative design using MidJourney

The program also emphasized practical learning through assignments, projects, research activities, and case studies. Students were encouraged to apply AI concepts to real-world scenarios, improving both technical and analytical skills.

Overall, the internship provided valuable exposure to industry-oriented AI practices and helped in building a strong foundation for future career opportunities in Artificial Intelligence and related domains.

Chapter 5

Learning Experiences

5.1 Knowledge Gained

- **Artificial Intelligence Fundamentals:** Gained a clear understanding of Artificial Intelligence concepts, including Machine Learning, Neural Networks, Reinforcement Learning, and AI applications in real-world scenarios.
- **Machine Learning Concepts:** Learned the basics of supervised and unsupervised learning, model training, prediction systems, and evaluation techniques used in AI projects.
- **Data Analysis and Processing:** Developed knowledge in data preprocessing techniques such as handling missing values, cleaning datasets, removing duplicates, and preparing data for analysis.
- **Academic Link:** The internship strengthened practical understanding of academic subjects such as Artificial Intelligence, Machine Learning, Data Analytics, Database Management Systems, and Software Engineering studied during the engineering course.

5.2 Technical Skills Developed

- **Programming Skills:** Improved proficiency in Python for implementing AI concepts, data handling, preprocessing, and Machine Learning-related activities.
- **Libraries and Frameworks:** Gained practical experience using Python libraries such as:
 - Pandas for data handling
 - NumPy for numerical computations
 - Matplotlib for data visualization
 - Scikit-learn for Machine Learning concepts
- **Development Tools:** Learned to work with tools such as:
 - Visual Studio Code (VS Code) for development
 - Git for version control and progress tracking
- **Level of Proficiency Achieved:** By the end of the internship, achieved beginner-to-intermediate level proficiency in AI-related programming, data analysis, visualization, and project implementation.

5.3 Soft Skills and Professional Development

- **Communication Skills:** Improved technical writing and communication skills through regular report preparation, documentation, and assignment submissions.
- **Time Management:** Learned to manage multiple assignments, project tasks, and deadlines efficiently throughout the internship period.
- **Problem-Solving Ability:** Developed the ability to identify problems, analyze possible solutions, and apply appropriate AI techniques to solve them effectively.
- **Analytical Thinking:** Improved logical and analytical thinking skills while working on data analysis tasks and AI-based projects.
- **Adaptability:** Adapted to new tools, technologies, and learning methods during the internship, improving flexibility and learning capability.
- **Professional Ethics:** Maintained discipline, consistency, proper documentation, and responsible use of AI tools throughout the internship activities.

5.4 Challenges Faced and Solutions

5.4.1.Challenge:Selecting Appropriate Primitive Polynomials

Identifying suitable primitive polynomials for generating Maximum Length Sequences was initially complex, as incorrect selection led to shorter periodic sequences and suboptimal performance.

Solution:

Studied finite field theory in detail and verified polynomial primitiveness using mathematical validation and simulation. Implemented automated checks in Python to confirm sequence period ($2^n - 1$) before finalizing the polynomial.

5.4.2.Challenge: Achieving Optimal Correlation Properties

Generated PRN sequences sometimes exhibited higher-than-expected cross-correlation.

Solution:

Performed systematic correlation analysis and adjusted LFSR tap configurations. Applied threshold-based filtering to select sequences meeting predefined correlation constraints.

5.4.3.Challenge: Extending Binary LFSR to $\mathbb{Z}_4$ Sequences

Transitioning from GF(2) binary systems to modulo-4 arithmetic introduced conceptual difficulties.

Solution:

Revisited modular arithmetic fundamentals and derived proper recurrence relations for $\mathbb{Z}_4$ systems. Carefully validated state transitions and balance properties through simulation.

5.4.4. Challenge: Debugging Algorithm Implementation

Minor logical errors in XOR operations and feedback connections led to incorrect sequence outputs.

Solution:

Adopted step-by-step debugging methods, manually verified intermediate states, and compared outputs with theoretical expectations to ensure correctness.

5.4.5.Challenge: Bridging Theory with Practical Application

Understanding how abstract mathematical concepts directly impact real GNSS signal performance required deeper analytical thinking.

Solution:

Connected mathematical results with physical interpretation, such as relating correlation peaks to signal acquisition. Regular discussions and performance testing through simulations strengthened practical understanding.

5.5 Practical Exposure

- **Real-world Application:** Applied theoretical knowledge of Artificial Intelligence and Machine Learning by working on practical assignments, AI tools, and the Student Stress Detector project.
- **Project Development Experience:** Gained hands-on exposure in developing an AI-based application that predicts student stress levels and provides supportive responses.
- **Industry Practices:** Observed how organizations provide structured learning, maintain documentation, manage assignments, and implement AI concepts in practical environments.
- During my internship at ISRO Telemetry, Tracking and Command Network (ISTRAC), I gained valuable practical exposure to real-time GNSS signal processing and sequence optimization techniques. I was involved in analyzing pseudo-random sequences generated using Linear Feedback Shift Registers (LFSR) and primitive polynomials over Galois Fields. This hands-on experience helped me understand how theoretical mathematical concepts are applied in real navigation systems. I worked on evaluating sequence properties such as autocorrelation, cross-correlation, balance, and linear complexity using computational tools and simulations.

- Exposure to performance metrics and signal quality assessment enhanced my analytical and problem-solving skills. I also observed how optimization techniques can be integrated with AI-based models to improve sequence selection efficiency in multi-satellite environments. Working in a professional research environment improved my documentation skills, technical communication, and structured analysis approach. This practical experience bridged the gap between academic knowledge and real-world GNSS applications.

Key Takeaways

- **Overall Summary:** The internship provided valuable exposure to Artificial Intelligence concepts, Machine Learning workflows, data analysis techniques, and AI-based tools through practical learning activities and projects.
- **Technical Growth:** Improved coding skills, analytical thinking and understanding of AI implementation through hands-on practice.
- **Personal Growth:** Developed confidence, discipline, consistency, and problem-solving skills while completing assignments and project activities.
- **Professional Growth:** Transitioned from theoretical understanding to practical implementation of AI concepts, improving readiness for future technical roles and industry opportunities.

Career Preparation: The internship strengthened interest in Artificial Intelligence and provided a strong foundation for pursuing future career opportunities in AI, Machine Learning, and Data Science fields

Chapter 6

Learning Outcomes

6.1 Skills Gained

During the internship, I gained both technical and professional skills that helped in improving my overall understanding of Artificial Intelligence and practical project development.

6.1.1 Technical Skills

- **GNSS and NavIC System Understanding.**

Developed a strong conceptual understanding of satellite navigation architecture, including space, control, and user segments, dual-frequency transmission (L5 and S-band), timing synchronization, and ionospheric delay correction mechanisms.

- **Sequence Generation and LFSR Design**

Gained hands-on experience in designing Linear Feedback Shift Registers (LFSRs) using Boolean logic and XOR operations. Implemented primitive polynomials over GF(2) to generate Maximum Length Sequences (m-sequences) with optimal periodic properties.

- **Non-Binary Sequence Modeling ($\mathbb{Z}_4$)**

Extended sequence generation concepts to modulo-4 arithmetic, analyzing balance properties, periodicity, and algebraic structures in $\mathbb{Z}_4$ systems.

- **Correlation and Performance Analysis**

Performed auto-correlation and cross-correlation analysis to evaluate sequence suitability for multi-satellite environments and signal acquisition reliability.

- **Programming and Simulation**

Implemented algorithms in Python for sequence generation, validation, and threshold-based optimization. Developed debugging, testing, and result interpretation skills.

- **Mathematical Modeling and Analysis**

Applied finite field mathematics, modular arithmetic, and recurrence relations to real-world GNSS signal processing problems.

6.1.1 Soft Skills

- **Communication Skills:** Improved report writing, documentation, and technical communication through regular internship activities.

- **Problem-Solving Skills:** Learned how to identify issues, analyze situations, and implement suitable solutions during project development and assignments.
- **Time Management:** Successfully handled deadlines, assignments, and project tasks systematically throughout the internship.
- **Analytical Thinking:** Improved logical thinking and decision-making abilities while working on AI-related tasks and projects.

I gained strong technical and analytical skills in GNSS sequence optimization using AI techniques. I developed a deeper understanding of LFSR design, primitive polynomials, and Galois Field theory for generating pseudo-random sequences. I improved my ability to analyze correlation properties and evaluate sequence performance metrics. Additionally, I enhanced my programming and simulation skills for testing optimization models. Working on real-time navigation problems strengthened my problem-solving ability, research mindset, and documentation skills, while also improving teamwork, technical communication, and structured report writing in a professional research environment.

5.6 Challenges Faced

During the internship, several challenges were encountered while learning new technologies and implementing practical concepts. These challenges helped in improving problem-solving abilities and learning adaptability.

Understanding AI Concepts

Initially, concepts such as neural networks, model prediction, and feature correlation were difficult to understand practically.

Solution:

Practiced multiple examples, referred to additional learning resources, and performed repeated hands-on exercises to improve conceptual clarity.

Data Visualization Difficulties

Selecting proper visualization methods and presenting data clearly using charts and graphs was initially confusing.

Solution: Experimented with different graph types and practiced visualization techniques using Matplotlib to improve clarity and representation.

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Version Control and Project Management

Managing project files and understanding Git workflows initially caused confusion during project tracking.

Solution:

Learned Git commands through practice and maintained proper version tracking and documentation for organized workflow management.

5.7 Results and Discussions

The internship provided practical exposure to Artificial Intelligence and Python Programming

5.7.1 Project Outcome

The internship projects in the Navigation Systems Area successfully integrated complex mathematical theory with practical implementation to optimize GNSS signal design. By utilizing Linear Feedback Shift Registers (LFSRs) and primitive polynomials, the project achieved efficient PRN code generation with maximum length periodicity ($2^n - 1$) for precise satellite identification. Technical validation was further strengthened through auto-correlation analysis for signal acquisition and cross-correlation optimization to minimize multi-satellite interference. Beyond binary sequences, the implementation of $\mathbb{Z}_4$ sequence modeling and the development of a threshold-based optimization process significantly improved code library efficiency. These efforts were encapsulated in a Python-based simulation framework that automates sequence verification, ultimately bridging the gap between digital logic and the high-precision requirements of satellite navigation systems.

5.7.2 Learning Results

The overall result of the internship projects was the successful design, implementation, and validation of optimized navigation signal sequences suitable for GNSS applications. PRN codes were generated using properly selected primitive polynomials, ensuring Maximum Length Sequences with correct periodicity and desirable statistical properties. Correlation analysis confirmed strong auto-correlation peaks for reliable signal acquisition and controlled cross-correlation values to minimize interference in multi-satellite environments. The implementation of $\mathbb{Z}_4$ sequence generation extended the work beyond traditional binary models, providing additional flexibility in sequence design and deeper insight into non-binary algebraic structures. Threshold-based filtering methods enabled systematic selection of high-performance sequences from larger candidate sets. All algorithms were implemented and verified through Python-based simulations, ensuring theoretical correctness and computational efficiency. Overall, the results demonstrated a successful integration of mathematical modeling, algorithm development, and performance evaluation, contributing to a structured and research-oriented approach to GNSS signal optimization.

5.8 Snapshots

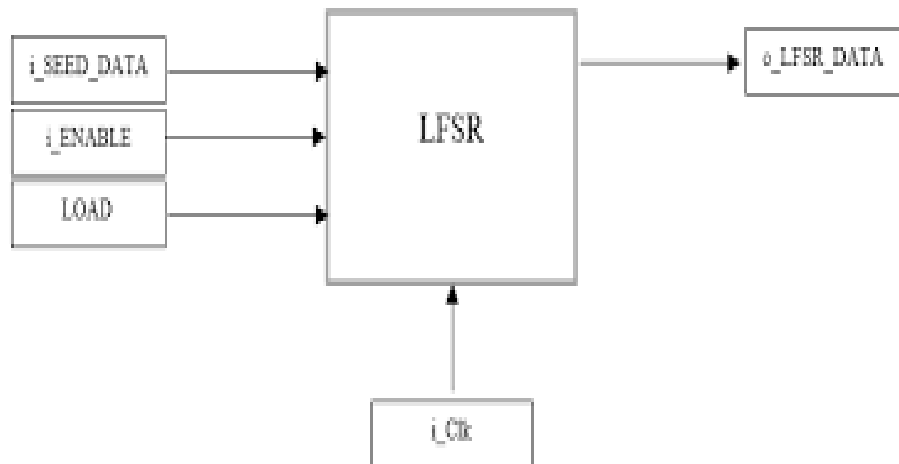


Fig.6.1 LFSR Block Diagram

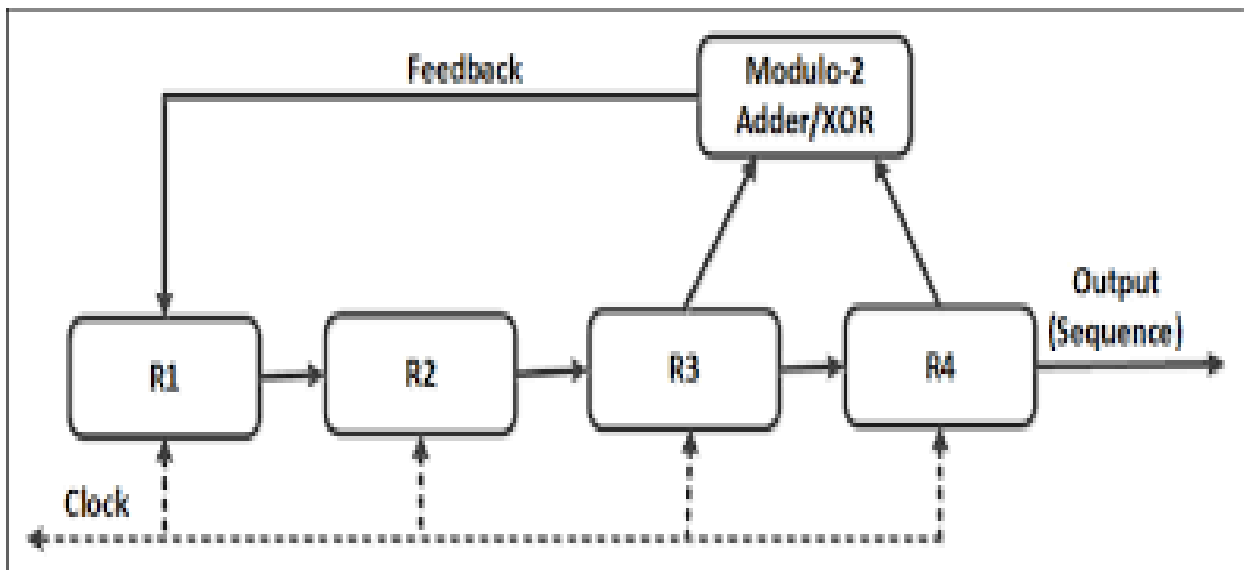


Fig 6.2 Sequence Generation using LFSR

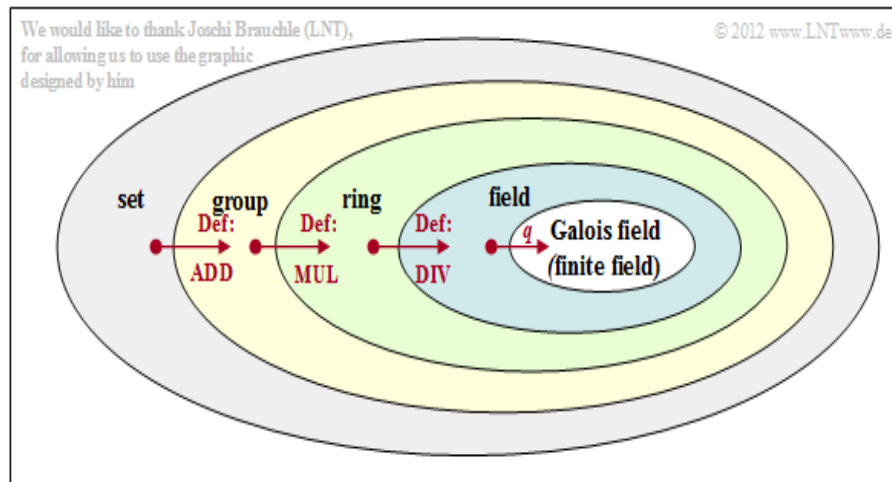


Fig 6.3 Galois Field

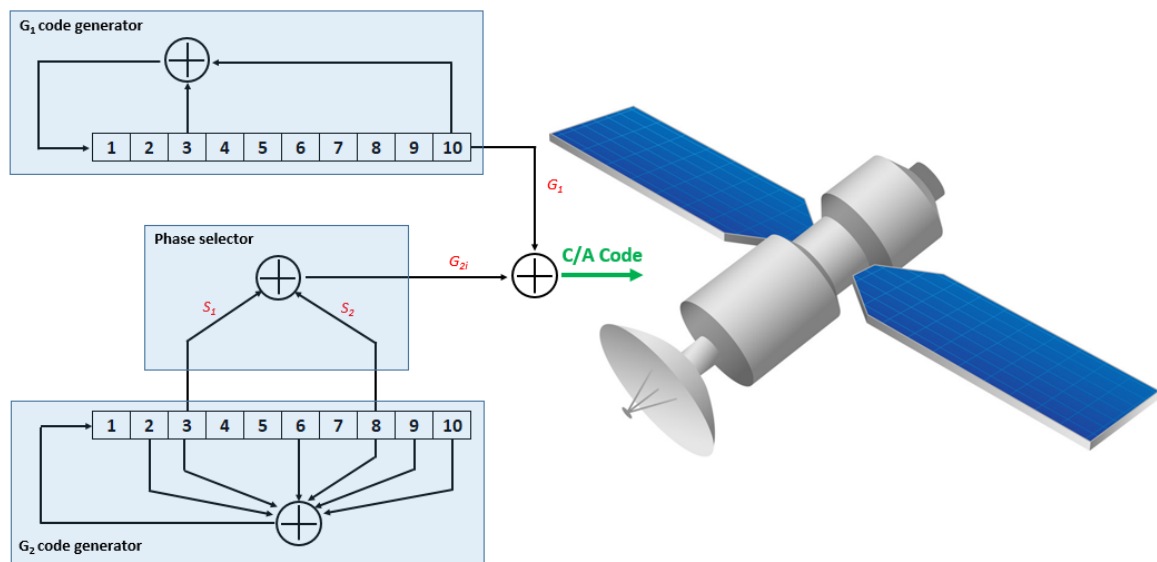


Fig 6.4 Satellite with PRN Code

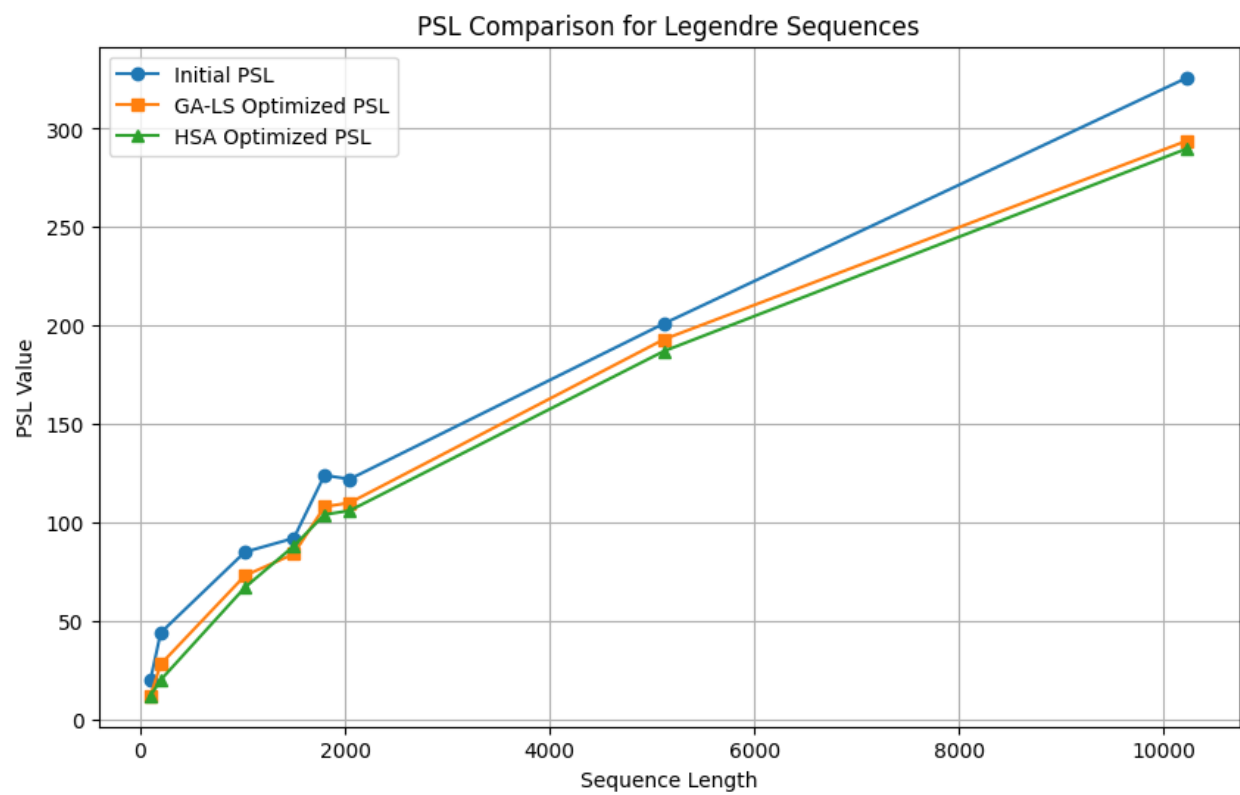


Fig 6.5 PSL Graph

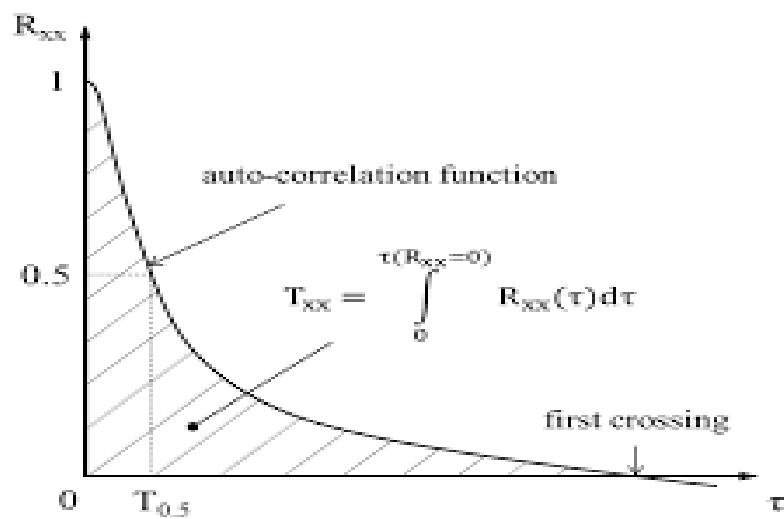


Fig 6.6 Auto-Correlation Function

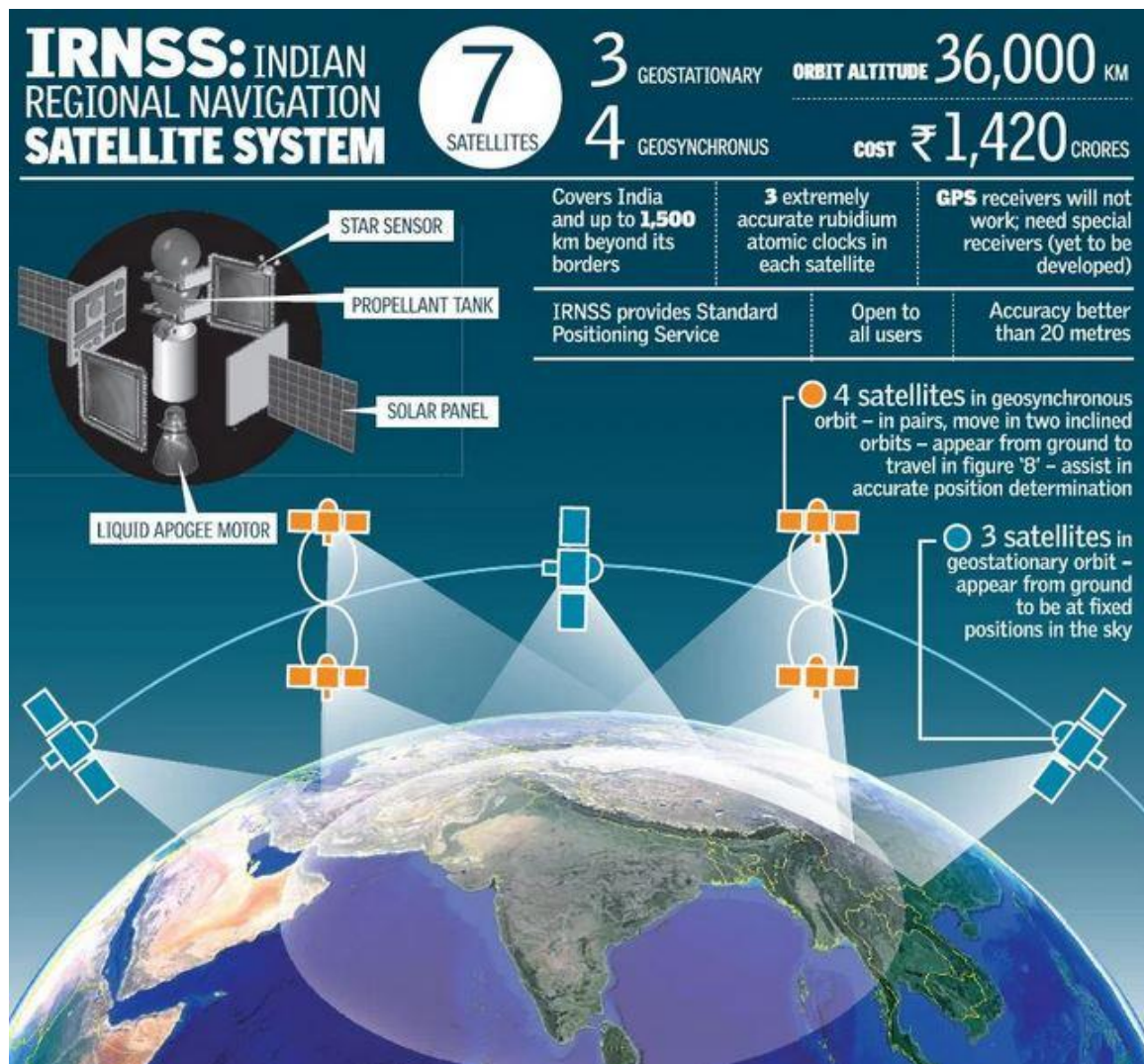


Fig 6.7 IRNSS



Fig 6.8 Working of NavIC

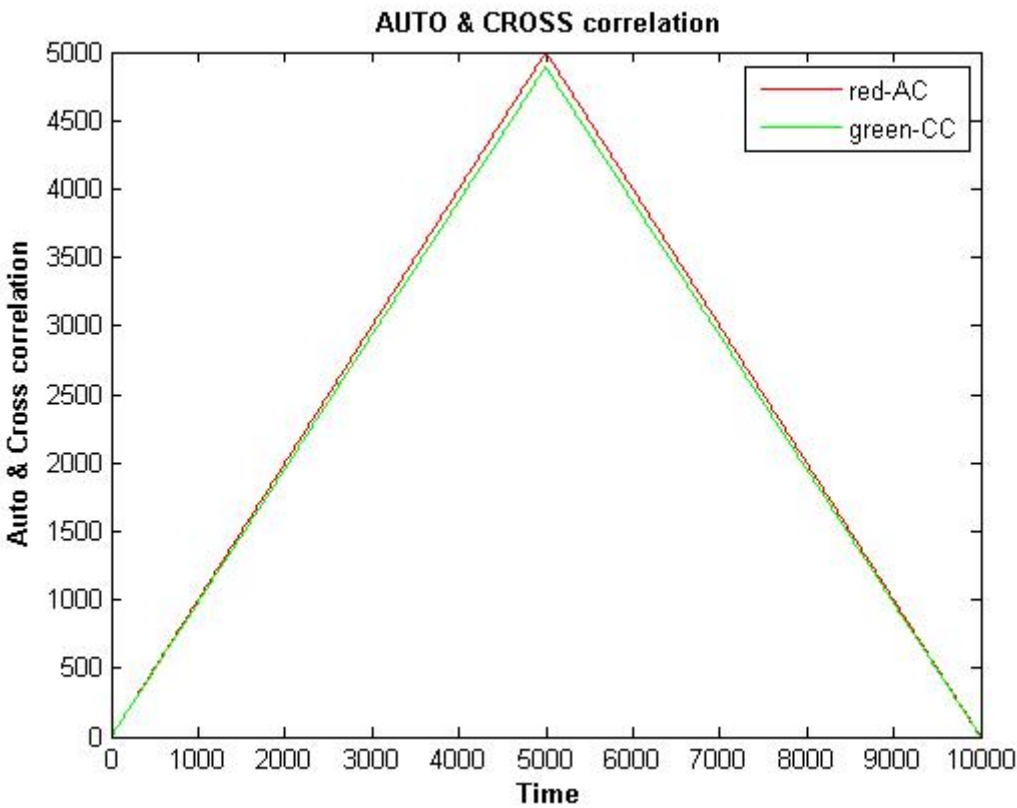


Fig 6.9 Peak Graph of ACR-CCR

5.9 Key Take Aways

- The internship provided valuable practical exposure to Artificial Intelligence, Machine Learning, and AI-based tools.
- Gained confidence in implementing theoretical concepts into real-world projects and applications.
- Improved both technical and professional skills through continuous learning and hands-on practice.
- Understood the importance of problem-solving, documentation, teamwork, and time management in professional environments.
- Developed a strong interest in Artificial Intelligence and related technologies, motivating further learning and career growth in this domain.
- The experience helped in preparing for future technical roles by improving industry readiness, adaptability, and practical understanding of AI workflows.

Chapter 7

Reflection and Analysis

7.1 Application of Academic Knowledge

The internship provided an opportunity to apply the theoretical concepts learned during academic studies to practical, real-world situations. Many subjects studied in the Computer Science and Engineering curriculum became easier to understand when implemented through practical activities and projects during the internship.

- **Artificial Intelligence and Machine Learning:** Concepts such as Machine Learning workflows, prediction systems, data preprocessing, and neural networks were practically explored while working on assignments and the Student Stress Detector project.
- **Programming Knowledge:** Python programming skills learned during academics were applied for implementing project functionalities, handling user input, preprocessing data, and integrating Machine Learning logic into the application.
- **Software Engineering Principles:** Concepts such as structured workflow, documentation, modular development, debugging, and testing were applied during project development activities.
- **Data Analysis and Visualization:** Academic understanding of data analysis and statistics was practically used while working with datasets, identifying patterns, and creating graphical visualizations for better interpretation.

The internship helped in bridging the gap between classroom learning and practical implementation by showing how theoretical knowledge is used in real technical environments.

7.2 Insights Gained

The internship at ISTRAC provided several valuable technical and professional insights that extended beyond classroom learning. One of the most important realizations was that satellite navigation is fundamentally a precision-driven system where timing accuracy, signal structure, and mathematical modeling are deeply interconnected. Even small variations in sequence design or synchronization can significantly affect positioning accuracy, highlighting the importance of rigorous validation and testing. From a technical standpoint, I gained insight into how abstract mathematical concepts such as finite fields, primitive polynomials, and recurrence relations directly influence real-world navigation signals. The implementation of LFSRs and PRN codes demonstrated how theoretical algebra is transformed into practical digital systems used in satellites. Working with $\mathbb{Z}_4$ sequences further revealed how non-binary structures can expand design flexibility and improve code diversity.

Another key insight was the importance of correlation properties in multi-satellite environments. Strong auto-correlation ensures accurate signal acquisition, while controlled cross-correlation prevents interference. This balance is crucial in GNSS system design. Professionally, I learned the importance of systematic problem-solving, structured documentation, and simulation-based verification. The experience reinforced that research requires patience, analytical thinking, and continuous refinement to achieve reliable and efficient solutions.

7.3 Areas for Improvement

While the internship was highly beneficial, it also highlighted areas for further growth. One key area is deepening expertise in advanced signal processing techniques, including modulation schemes, synchronization algorithms, and receiver design. Expanding knowledge in hardware implementation, such as FPGA-based sequence generation and real-time embedded systems, would strengthen practical application skills.

Another area for improvement is enhancing proficiency in advanced mathematical tools and simulation platforms beyond Python, such as MATLAB or specialized GNSS software environments. Strengthening understanding of real-time system constraints and performance optimization under hardware limitations would also be valuable.

Additionally, improving technical documentation skills, presentation abilities, and research publication writing can further support academic and professional advancement. Developing interdisciplinary knowledge in communication theory, coding theory, and satellite system design will help build a more holistic understanding of navigation systems.

By focusing on these improvement areas, future contributions to GNSS research and satellite technology development can be made more impactful and technically robust.

7.4 Career Impact

The internship at ISTRAC (Navigation Systems Area) has had a significant impact on my career direction and professional aspirations. signal processing, and applied mathematics. Working on PRN code generation, LFSR modeling, $\mathbb{Z}_4$ sequence design, and correlation analysis helped bridge the gap between theoretical coursework and practical engineering applications. From a technical perspective, the internship enhanced my understanding of finite field mathematics, digital sequence

generation, and algorithm validation through simulation. Implementing models in Python improved programming confidence and problem-solving skills. The experience also developed research-oriented thinking, including systematic experimentation, analytical evaluation, and performance optimization.

Professionally, the internship improved my ability to work in a structured research environment, follow documentation standards, and communicate technical concepts clearly. It strengthened discipline, attention to detail, and logical reasoning—skills essential for careers in space research, GNSS development, communication systems, or advanced software engineering roles. Overall, the internship has positively shaped my career path by motivating me to pursue advanced research or technical roles in satellite navigation, signal processing, and emerging space technologies

Chapter 8

Conclusion

8.1 Summary of Internship Experience

The internship at ISTRAC within the Navigation Systems Area (NSA) was a highly enriching and technically intensive experience that provided deep exposure to satellite navigation systems, particularly NavIC. The internship followed a structured progression from foundational research to mathematical modeling and practical implementation. Initially, significant emphasis was placed on understanding the architecture of NavIC, including its space, control, and user segments, along with signal structure, timing synchronization, and dual-frequency transmission principles. This domain study built a strong conceptual base in GNSS fundamentals, error sources, and correction mechanisms such as ionospheric delay mitigation.

The technical phase focused on PRN code generation, LFSR implementation, primitive polynomial selection, and sequence optimization. Advanced sequence modeling such as $\mathbb{Z}_4$ -based sequence generation expanded the scope beyond binary systems. Threshold-based sequence evaluation and correlation analysis ensured practical validation of theoretical models. Python simulations were extensively used to test algorithm performance and verify sequence properties. Overall, the internship successfully integrated theoretical knowledge with real-world GNSS signal processing challenges, providing comprehensive exposure to research-oriented problem solving in navigation systems.

8.2 Professional and Academic Growth

The internship significantly enhanced both professional competence and academic understanding. Academically, it strengthened knowledge in finite field mathematics, Galois theory, digital logic design, and correlation analysis. Concepts such as LFSRs, primitive polynomials, maximum length sequences, and non-binary algebra were not only studied theoretically but also implemented practically, deepening conceptual clarity. Exposure to GNSS architecture and dual-frequency systems improved understanding of real-world engineering applications of mathematical models. Professionally, the internship improved analytical thinking, problem-solving ability, and research methodology. Working on sequence optimization required systematic experimentation, performance evaluation, and validation through simulation. Coding implementations in Python enhanced programming proficiency and debugging skills. The experience also fostered discipline in documentation, structured reporting, and technical communication. Additionally,

exposure to a real research environment at ISTRAC cultivated teamwork, adaptability, and a deeper appreciation for precision-driven engineering systems. Overall, the internship played a crucial role in bridging academic learning with industry-level research and technological applications.

8.3 Final Conclusion

The internship at ISTRAC in the Navigation Systems Area (NSA) was a transformative academic and professional experience that provided in-depth exposure to satellite navigation technologies, particularly NavIC. The structured progression from domain research to technical implementation enabled a comprehensive understanding of GNSS architecture, signal design, and performance optimization. Studying the three-segment framework of NavIC and its dual-frequency transmission strategy strengthened conceptual clarity on positioning accuracy, timing synchronization, and atmospheric error mitigation.

The practical implementation of PRN code generation, LFSR modeling, primitive polynomial selection, and $\mathbb{Z}_4$ sequence design bridged theoretical mathematics with real-world navigation applications. Correlation analysis and threshold-based validation further reinforced the importance of sequence optimization in multi-satellite environments. Python-based simulations ensured that theoretical models were computationally verified and performance-efficient.

Overall, the internship successfully integrated mathematical modeling, algorithm development, and system-level understanding into a cohesive learning experience. It enhanced technical competence, analytical thinking, and research methodology while fostering a deeper appreciation for precision engineering in satellite navigation systems. This experience has laid a strong foundation for future research and advanced work in GNSS and communication systems.

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Institute Vision

To become a recognized technical education center with a global perspective.

Institute Mission

To provide learning opportunities that foster students' ethical values, intelligent development in science technology and social responsibility so that they become sensible and contributing members of society.

Department Vision

To become a Centre of excellence producing “Creators of Innovative Technology” who can contribute positively to the ever changing industrial demands and societal needs.

Department Mission

M1: To encourage participation of faculty and students in research activities for enhancing their subject knowledge and acquire information regarding current trends

M2: To provide exposure to students on latest tools and technologies in area of Computer Science and Engineering

M3: Preparation of our graduates for leadership in profession and in higher education by providing excellent teaching learning environment enabling them to serve the society